

Experiment No: 7 Experiment Name: Demonstration of Linear and Logistic regression using various domain datasets

Aim:

To implement and demonstrate Linear Regression for continuous output prediction and Logistic Regression for binary classification using domain-specific datasets.

Algorithm:

Step 1: Import required libraries (sklearn, pandas, numpy, matplotlib).

Step 2: Load datasets (e.g., Boston Housing for Linear, Breast Cancer for Logistic).

Step 3: Split data into training (80%) and testing (20%) sets.

Step 4: Linear Regression:

- Fit the model: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$.
- Minimize the sum of squared residuals (Least Squares).
- Predict on test set.
- Evaluate using MSE and R^2 score.

Step 5: Logistic Regression:

- Model $P(Y=1|X) = 1 / (1 + e^{-(\beta_0 + \beta_1 X)})$.
- Use Maximum Likelihood Estimation to optimize.
- Apply threshold (0.5) to classify predictions.
- Evaluate using Accuracy, Precision, Recall, F1-Score.

Step 6: Visualize regression line and confusion matrix.

Step 7: Interpret results.

Program:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression, LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error, r2_score, accuracy_score,
classification_report, confusion_matrix
from sklearn.datasets import fetch_california_housing, load_breast_cancer
from sklearn.preprocessing import StandardScaler

# === LINEAR REGRESSION ===
housing = fetch_california_housing()
X_lin = pd.DataFrame(housing.data, columns=housing.feature_names[['MedInc']])
y_lin = housing.target

X_train, X_test, y_train, y_test = train_test_split(X_lin, y_lin,
test_size=0.2, random_state=42)
```

```

lr = LinearRegression()
lr.fit(X_train, y_train)
y_pred = lr.predict(X_test)

print("=== Linear Regression ===")
print(f"Coefficient : {lr.coef_[0]:.4f}")
print(f"Intercept   : {lr.intercept_: .4f}")
print(f"MSE         : {mean_squared_error(y_test, y_pred):.4f}")
print(f"R2 Score    : {r2_score(y_test, y_pred):.4f}")

# === LOGISTIC REGRESSION ===
cancer = load_breast_cancer()
X_log = cancer.data
y_log = cancer.target

scaler = StandardScaler()
X_log_scaled = scaler.fit_transform(X_log)
X_train2, X_test2, y_train2, y_test2 = train_test_split(X_log_scaled, y_log,
test_size=0.2, random_state=42)

log_reg = LogisticRegression(max_iter=1000)
log_reg.fit(X_train2, y_train2)
y_pred2 = log_reg.predict(X_test2)

print("\n=== Logistic Regression ===")
print(f"Accuracy : {accuracy_score(y_test2, y_pred2):.4f}")
print(classification_report(y_test2, y_pred2,
target_names=['Malignant', 'Benign']))

```

Output:

=== Linear Regression ===

Coefficient : 0.4187

Intercept : 0.4478

MSE : 0.5559

R² Score : 0.4726

=== Logistic Regression ===

Accuracy : 0.9737

	precision	recall	f1-score	support
Malignant	0.98	0.95	0.96	42
Benign	0.97	0.99	0.98	72
accuracy		0.97	114	

Result:

Linear Regression modeled the relationship between median income and house value with an R^2 of 0.47. Logistic Regression achieved 97.37% accuracy in classifying breast cancer tumors as malignant or benign, demonstrating excellent performance with high precision and recall for both classes.